

COMMUTATIVE ALGEBRA EXAM

1. HILBERT'S NULLSTELLENSATZ AND ZARISKI TOPOLOGY

Exercise 1.1. Let $R = \mathbb{R}[t]$ and X be its maximal spectrum. Prove that X with its Zariski topology is homeomorphic to the maximal spectrum of $\mathbb{C}[t]$.

Proof. It's clear that the maximal spectrums of $\mathbb{R}[t]$ and $\mathbb{C}[t]$ have the same cardinality. Choose any bijection between them. We claim this is a homeomorphism by being a closed map. This follows since closed sets in any maximal spectrum are the finite sets (and the image of a finite set under a bijection is finite).

To check a closed set F is finite, suppose $\mathfrak{m} \in F$. The basis of closed sets is given by the complements of $U(f)$, which are defined by $\mathfrak{m}' \in U(f) \iff f \notin \mathfrak{m}'$. Thus for the closed set $F = U(f)^c$ we have that $\mathfrak{m} \in F \iff f \in \mathfrak{m}$. Now since $K[t]$ is a PID, we may assume $\mathfrak{m} = (g)$, so $f \in \mathfrak{m}$ becomes g is a factor of f . But f can have only finitely many factors.

To check any finite set is closed, we only use the maximal ideals $t - a$ for any $a \in \mathbb{R}$ or $\mathbb{C}$. $\square$

Exercise 1.2. Let $P(x, y)$ be an irreducible polynomial and $R = \frac{\mathbb{C}[x, y]}{(P)}$. Prove that all ideals in R are principal, or find a counterexample.

Proof. This is false. Consider $R = \mathbb{C}[x, y]/(y - x^2)$ and the ideal $(x, y) \subset \mathbb{C}[x, y]$. This ideal contains $y^3 - x^2$, so it's an ideal of R , but it is not principal! Indeed, even after substituting $y^3 = x^2$, there's no way to obtain y from x or vice versa. [Proof can be improved] $\square$

2. NOETHERIAN RINGS

Exercise 2.1. Let M be an A -module, where A is not necessarily Noetherian, and $N_1, N_2 \subset M$ submodules which satisfy $N_1 \cap N_2 = 0$. Assume that the modules M/N_1 and M/N_2 are Noetherian. Prove that M is Noetherian.

Failed proof. Let N be a submodule (=ideal) of M . Projecting N to M/N_1 we obtain a finitely generated module. Suppose the generators are $x_i + N_1$, where $x_i \in N$. The fact that they are generators means that for any $x \in N$ we have $x + N_1 = \sum a_i x_i + N_1$, that is, $x - \sum a_i x_i \in N_1$. The same goes for N_2 , say, $x - \sum b_i x'_i \in N_2$ for any $x \in N$. But then $\sum a_i x_i - \sum b_i x'_i \in N_1 \cap N_2$, a contradiction (no...) $\square$

Exercise 2.2. Let A be a ring without zero divisors. Suppose that any descending chain of ideals in A stabilizes. Prove that A is a field.

Proof. Suppose that A is not a field and let's show that there's a descending chain of ideals which does not stabilize. Let a be not invertible (since A is not a field). Then

$$(a) \subset (a^2) \subset (a^3) \subset \dots$$

does not stabilize. Indeed, if $(a^n) = (a^{n+1})$ we obtain $a^n = ba^{n+1} \iff a^n(1-ba) = 0 \implies 1 = ba$, since A is a domain, which is a contradiction because a is not invertible. $\square$

3. IRREDUCIBLE POLYNOMIALS AND GALOIS THEORY

Exercise 3.1. Let $k = \mathbb{C}(t)$ be the field of rational functions on $\mathbb{C}$. Prove that for any n there exists an irreducible polynomial $P(t) \in k[t]$ of degree n .

Proof. Let $P(x) \in k[x] = \mathbb{C}(t)[x]$ be $P(x) = x^n - t$. This polynomial has no root in $k = \mathbb{C}(t)$, for if $(f/g)^n = t$ then $n(\deg f - \deg g) = 1$, which is impossible for $n > 1$. But unfortunately this doesn't prove that it's irreducible: in $\mathbb{R}[x]$, the polynomial $(x^2 + 1)(x^2 + 2) = x^4 + 3x^2 + 2$ is reducible and has no roots.

To prove $x^n + t$ is irreducible we shall use the Gauss lemma. Forget the above notation and suppose that $fg = x^n + t$, where $f, g \in \mathbb{C}(t)[x]$. We want to clear the denominators in the coefficients of f and g , so we multiply by the product D of all such denominators and obtain

$$f_1 g_1 = (x^n + t)D,$$

where $D \in \mathbb{C}[t]$ and $f_1, g_1 \in \mathbb{C}[t][x]$. Now we apply Gauss lemma, that is, the content of the product is the product of the contents. Since $x^n + t$ is primitive (i.e. has content 1), then the content of $f_1 g_1$ equals the content of D , but D is a constant (constants here are polynomials in $\mathbb{C}[t]$). We obtain

$$FG = x^n + t$$

where F and G are the primitive parts of f_1 and g_1 . This holds up to multiplication by a complex number.

Now we reduce modulo t and obtain

$$[F][G] = [x^n],$$

and conclude that, as classes,

$$[F] = [cx^m], \quad [G] = [c^{-1}x^{n-m}].$$

Thus we see that each of F, G lost their constant term when we reduced, i.e. that their constant terms are divisible by t . This means that t^2 divides the constant term of the product FG . But this is impossible, since the constant term of FG is t .

The only way this makes sense is if $m = 0$ or $m = n$. To conclude we note that reduction modulo t preserves the degree of both F , and G because their leading coefficients multiply to 1. Thus $m = 0$ or $m = n$ imply that F or G is a unit, which in turn implies so for f or g . $\square$

Exercise 3.2. Prove that for any $d \in \mathbb{Z}^{>0}$ there exists an irreducible polynomial $P \in \mathbb{C}[x, y]$ such that $P(x, y) = 0$ defines a smooth submanifold in $\mathbb{C}^2$.

Proof. $x^n + y$ since $\partial_y = 1$. This is an irreducible polynomial because y is irreducible. More precisely, if $x^n + y = fg$, then either of f, g must have no y variable, say $f \in \mathbb{C}[x]$, and the other one must be degree 1 in y . Suppose $g = yg' + g''$ for $g', g'' \in \mathbb{C}[x]$. But then $x^n + y = fg = yfg' + fg''$, which means that f cannot have positive degree in x , but $f \in \mathbb{C}[x]$, so it is a unit. $\square$

4. TENSOR PRODUCT OF R-MODULES

Exercise 4.1. Let A be a ring without zero divisors, and M an A -module which is flat over all local rings $A_{\mathfrak{p}} \supset A$ obtained by localizing A at prime ideals. Prove that M is flat.

Proof. Let

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

be exact. Then consider

$$0 \longrightarrow M_1 \otimes_A M \xrightarrow{\varphi} M_2 \otimes_A M \longrightarrow M_3 \otimes_A M \longrightarrow 0.$$

We need to show that the map $\varphi : M_1 \otimes_A M \rightarrow M_2 \otimes_A M$ is injective. So basically the claim is: if that map is injective over all localizations $A_{\mathfrak{p}}$ then it is injective over A .

So it's basically two steps: (1) show that

$$(4.1.1) \quad (\text{Ker } \varphi)_{\mathfrak{p}} = \text{Ker } \varphi^{\mathfrak{p}}$$

where $\varphi^{\mathfrak{p}}$ is the induced map over $A_{\mathfrak{p}}$ for any $\mathfrak{p} \subset A$ prime. More precisely:

$$0 \longrightarrow (M_1)_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}} \xrightarrow{\varphi^{\mathfrak{p}}} (M_2)_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}} \longrightarrow (M_3)_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}} \longrightarrow 0$$

where the localizations of each module are given by tensoring with $A_{\mathfrak{p}}$ over A , e.g. $M_{\mathfrak{p}} = A_{\mathfrak{p}} \otimes_A M$ by definition. Thus the above sequence is exactly

$$0 \longrightarrow (M_1 \otimes_A M)_{\mathfrak{p}} \xrightarrow{\varphi^{\mathfrak{p}}} (M_2 \otimes_A M)_{\mathfrak{p}} \longrightarrow (M_3 \otimes_A M)_{\mathfrak{p}} \longrightarrow 0$$

and we see that Equation 4.1.1 follows by definition. Indeed, both sets are barely

$$\left\{ \frac{a}{b} x : \varphi(x) = 0, \frac{a}{b} \in A_{\mathfrak{p}} \right\}.$$

By the exercise's hypothesis, we know that $\text{Ker } \varphi^{\mathfrak{p}} = 0$.

Then, step (2) is: show that if any A -module K is zero as an $A_{\mathfrak{p}}$ -module for any $\mathfrak{p} \subset A$, then it is zero as an A -module. Here's AI's proof: consider the annihilator of an element $m \in M$, it's the set of scalars that kill it, i.e. $\text{Ann}(m) = \{a \in A : am = 0\}$. It's nonempty by hypothesis (yes, the fact that m is zero over $A_{\mathfrak{p}}$ literally means that there is $a \notin \mathfrak{p}$ such that $am = 0$). This just says that $\text{Ann}(m) \not\subset \mathfrak{p}$ for any $\mathfrak{p}$. But that's not true: any ideal is contained in some maximal (and thus prime) ideal. The reason for that is Zorn's lemma: the set of ideals containing $\text{Ann}(m)$ is inductive (i.e. every chain is bounded, in this case by the union of the ideals), so it has a maximal element, and maximal ideals are prime because the quotient is a field, which is in particular a domain. □

Definition 4.2. A *boolean ring* is a ring such that all its elements are idempotents, that is, satisfy $a^2 = a$.

Exercise 4.3. Prove that any finitely generated module over a Boolean ring is flat.

Failed proof. The claim is false. Consider the boolean ring $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, equipped with the usual ring structure for direct sums which is given by applying the operations entry by entry.

This module has torsion:

$$([1], [0])([0], [1]) = ([0], [0]).$$

Ah! But this ring is not a domain. Forget it. $\square$

5. TENSOR PRODUCT OF RINGS

Exercise 5.1. Let R_1, R_2 be rings over $\mathbb{C}$ such that $R_1 \otimes_{\mathbb{C}} R_2$ is finitely generated. Prove that R_1, R_2 are finitely generated, or find a counterexample.

Proof. Suppose $R_1 \otimes_{\mathbb{C}} R_2 = \mathbb{C}[z_1, \dots, z_n]$ and $z_j = \sum_k a_{jk} \otimes b_{jk}$. Let $A \subset R_1$ be the algebra generated by the a_{jk} . It's clear that the generators z_j are in $A \otimes_{\mathbb{C}} R_2$. Thus, any polynomial on the z_j lies in $A \otimes_{\mathbb{C}} R_2$. That is, $R_1 \otimes_{\mathbb{C}} R_2 = A \otimes_{\mathbb{C}} R_2$.

Let $\lambda : R_2 \rightarrow \mathbb{C}$ be a linear functional with $\lambda(1) = 1$. Consider the operator $\text{id} \otimes \lambda : A \otimes_{\mathbb{C}} R_2 \rightarrow A$.

Now take $r_1 \in R_1$. Then $r_1 \otimes 1 \in R_1 \otimes_{\mathbb{C}} R_2 = A \otimes_{\mathbb{C}} R_2$. Therefore $r_1 = \text{id} \otimes \lambda(r_1 \otimes 1) \in A$, and A is finitely generated. $\square$

Exercise 5.2. Let A be the ring of complex-analytic functions on $\mathbb{C}$, and $k(A)$ its fraction field. Prove that the transcendence degree of A over $\mathbb{C}$ is infinite.

6. NORMALITY, IRREDUCIBILITY, SMOOTHNESS

Exercise 6.1. Prove that the ring $\mathbb{C}[x, y, z]/(y^2 - xz)$ is integrally closed.

Proof. Consider a parametrization of the corresponding affine surface:

$$\varphi : (t, s) \mapsto (t^2, ts, s^2)$$

This gives an isomorphism

$$\frac{k[x, y, z]}{(y^2 - xz)} \cong k[t^2, ts, s^2] \subset k[t, s].$$

To check this we need to prove that its kernel is indeed $(y^2 - xz)$. For this let $F \in \text{Ker } \varphi$. Consider F as a polynomial in y and use division algorithm by $y^2 - xz$ to obtain:

$$F = Q(y^2 - xz) + A(x, z) + yB(x, z).$$

since the residue must have degree in y strictly less than $2 = \deg(y^2 - xz)$. Applying φ we obtain

$$0 = A(s^2, t^2) + stB(s^2, t^2),$$

which have different parity degrees on s and t , meaning both summands are zero. The fact that $(y^2 - xz) \subset \text{Ker } \varphi$ is obvious, so we have equality as desired.

Let $A = \mathbb{C}[t^2, ts, s^2]$. To prove A is integrally closed notice that A is exactly the subring of $\mathbb{C}[t, s]$ of polynomials which remain invariant under $\varphi : (t, s) \mapsto (-t, -s)$. Indeed, it is clear that elements A are φ -invariant, and conversely, any φ -invariant polynomial must have even degree in all variables and thus it must be in A .

Now let $u \in \text{Frac } A$ be the root of a monic polynomial with coefficients in A . Since $A \subset \mathbb{C}[t, s]$, and $\mathbb{C}[t, s]$ is integrally closed, we conclude $u \in \mathbb{C}[t, s]$. But since u is a quotient of elements in A , it is φ -invariant. Therefore $u \in A$. $\square$

Exercise 6.2. Let $\zeta_n \in \mathbb{C}$ be a primitive root of unity of degree n . Define the action of the cyclic group $G = \mathbb{Z}/\{0\}Z$ on $\mathbb{C}^2$ with coordinates x, y in such a way that the generator t maps x to $\zeta_n x$ and y to $\zeta_n^{-1} y$. Prove that the quotient $\mathbb{C}^2/G$ is singular.

Proof. The ring of invariants is $\mathbb{C}[x^n, y^n, xy]$. Consider the ideal $\mathfrak{m} = (x^n, y^n, xy)$. It is clear that, for $n \geq 2$, the cotangent space $\mathfrak{m}/\mathfrak{m}^2$ has three linearly independent vectors, namely, the classes of x^n, y^n and xy , and thus it can't be regular. $\square$